Hash Tables

- COMP26120 - Week 3

- Hashing, collisions, load factor, resizing

Hash functions

- Map a key to an array index
- h(k) = k mod m
- Spread keys uniformly across buckets

Collisions

- Two keys can map to the same index
- Separate chaining: a list per bucket
- Open addressing: probe for a free slot

Load factor

- α = n / m (items / buckets)
- Keep α below 0.75
- Lookup is O(1) on average

Resizing

- Double the array when the load factor is too high
- Rehash every key
- Insert is amortised O(1)